No of Pages :4 Course Code : 23MX13

Roll No:
(To be filled in by the candidate)

PSG COLLEGE OF TECHNOLOGY, COIMBATORE - 641 004

SEMESTER EXAMINATIONS, DECEMBER 2024

MCA Semester : 1

23MX13 DATA STRUCTURES

Time : 3 Hours **Maximum Marks :100**

INSTRUCTIONS:

1. Answer **ALL** questions. Each question carries 25 Marks.
2. Course Outcome : Table

Qn.1	CO1	Qn.2	CO2	Qn.3	CO3	Qn.4	CO4

COs	At the end of learning this course, students will be able to
CO1	Analyse the efficiency and complexity of iterative and recursive algorithms.
CO2	Formulate algorithms using linear data structures to solve problems and analyze their complexity.
CO3	Formulate algorithms using non-linear data structures to solve problems and analyze their complexity.
CO4	Implement searching and sorting algorithms and choose the most suitable one for specific scenario.

Question **(BTL)**

1. a) i) Suggest a suitable data structure for the following scenario and write the corresponding ADT. (L2)

Imagine you are working on a ticket booking system for a movie theatre. When customers request tickets, requests are to be processed ensuring fairness. (3)

ii) For a particular task, different algorithms are written and their time complexities are O(nlogn), O(n), O(log n), O(2^n)and O(n^2). Based on complexity, which of these algorithms is good and which is worst? Justify your answer. (2) (L2)

b) i) Write an algorithm to generate the Fibonacci sequence up to a given number of terms. For instance, if input is 10, following sequence is to be displayed. (L3)

0, 1, 1, 2, 3, 5, 8, 13, 21, 34.

Show that the algorithm possess all the properties of an algorithm. (5)

ii) Indicate the complexity of the following code segments using Big O notation. (3) (L3)

```
1) for ( j = n/2; j ≤ n; j++)
      for( k =1; k ≤ n; k = 2 * k)
         for (m =1; m ≤ n ; m = m *2)
               printf("#")
```

2)
```
int fun(int n){
    int count = 0;
    for (int i = n; i > 0; i /= 2)
        for (int j = 0; j < i; j++)
            count += 1;
    return count;    }
```

3)
```
function (n)  {
for(int i =1; i < =n ; i + +)
  for (int j = 1 ; j <= n ; j+ = i )
      printf (" *") ;      }
```

c) i) Write the recurrence relation describing the following algorithm. Assume the complexity of the function Merge(int, int, int) is O(n). Solve the recurrence relation and indicate the complexity of the code in Big Oh notation. (6) (L4)

```
Algorithm Mergesort(low, high) {
    If (low <high) {
    Mid = (low +high) /2
    Mergesort(low, mid)
    Mergesort(mid+1, high)
    Merge(low, mid, high)
}
```

ii) Define Master theorem. Find the time complexity for the following recurrence relation using master theorem. (6) (L3)

i. $T(n) = 4T(n/2) + n^2$

ii. $T(n) = \sqrt{2}\, T(n/2) + \log n$

2. a) What are the various ways of representing two dimensional arrays in storage? Give an example for each. Suggest an efficient way of storing the tri diagonal matrix and write the corresponding addressing function. A tri diagonal matrix of order 4 x 4 is shown below: (5) (L2)

x	x	0	0
x	x	x	0
0	x	x	x
0	0	x	x

b) Write an algorithm to (i) convert infix expression to postfix form and Trace these algorithm on (((a+ b) * (a*a))+ c) (ii) count the numbers of elements in the given stack (5+3) (L3)

c) Formulate algorithm to perform the following: (12) (L3)

i) Given two singly linked lists, create a new circularly linked list with the even elements of the given lists in order.

ii) Merge two singly linked lists (in sorted order)

(OR)

c) i) Write algorithms for the basic operations on linked queue. (6) (L4)

ii) Given a queue of integers, how do you reverse the order of the elements of the queue? (6)

Example :

Input Queue [10, 20, 30, 40, 50, 60, 70, 80, 90]

Output: Queue [90, 80, 70, 60, 50, 40, 30, 20, 10]

3. a) i) Write an algorithm to count the number of leaf nodes in the given binary tree. (3) (L3)

ii) Draw i. a strictly binary tree, but not a complete binary tree, whose postorder traversal sequence is ABCDEFG and ii. a complete binary tree whose preorder sequence is A B C D E F G (2) (L3)

b) Consider the tree given in Fig. 1, Show the tree after inserting 45 and 25 and after deleting 8, 12, 6 and 10.Formulatean algorithm to perform deletion in Binary Search Tree. (8) (L4)

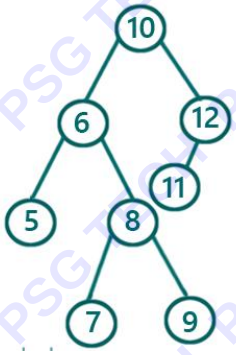

Fig. 1

c) List the nodes of the graph shown in Fig. 2 when traversed using i) Breadth First Search and ii) Depth First Search. Write the algorithm for DFS. Formulate an algorithm to check whether the graph is connected. Trace the algorithms with the graph shown in Fig.2. (12) (L5)

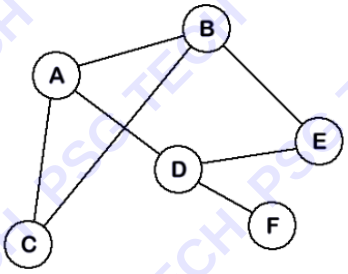

Fig.2

4. a) Illustrate Heap sort with the following elements: 34, 67, 89, 12, 45, 65, 90. What is the complexity of Heap Sort? (5) (L1)

b) Formulate binary search algorithm. Trace the algorithm on a sample input. Indicate the best and worst case complexity (8) (L3)

c) i) What is Hashing? What is collision? Explain any three methods of resolving collisions. Consider a hash table of size 11 with hash function h(k)=k mod 11. Draw the table that results after inserting, the following values in the given order: 9, 26, 50, 15, 2, 21, 36, 22 and 31. Resolve collisions using each of the collision resolving techniques discussed. Describe insertion, searching and deletion in each of the methods discussed. Comment on the effectiveness of the collision resolution techniques. (12) (L4)

(OR)

ii) 1) Write a sorting algorithm with minimum number of swapping of elements. What is the complexity of the algorithm? Is the algorithm stable? Justify your answer. (L3)

Apply the algorithm to sort the following list of elements:12, 25, 3, 45, 56, 34 and 70. Show the results of the intermediate passes. Analyze the best and worst case complexity. (6)

2) Illustrate Radix Sort with the following elements: 45, 23, 67, 12, 30, 50, 5. Compare the performance of Radix Sort with Insertion sort. (6) (L2)

CO : COURSE OUTCOME; BTL : BLOOM'S TAXONOMY LEVEL

/END/

FD/RL